EXPERIMENT : 5

Aim: To understand Continuous Integration, install Jenkins, and build CI/CD pipelines using Declarative and Scripted syntax.

LO mapped: **LO3**

LO3: To Build and Manage automated CI/CD pipeline, for application development, packaging, testing and deployment

Theory:

Continuous Integration (CI)

Continuous Integration (CI) is a software development practice where developers frequently integrate their code changes into a common repository. Every integration automatically initiates processes such as compiling the application and executing tests. This enables early detection of errors, minimizes integration conflicts, enhances collaboration among developers, and helps maintain the overall quality and stability of the software throughout the development lifecycle.

Jenkins

Jenkins is an open-source automation server widely used for implementing Continuous Integration and Continuous Delivery (CI/CD). It automates routine software development tasks, including source code compilation, testing, report generation, and application deployment. Jenkins supports a large collection of plugins, allowing seamless integration with various development tools, version control systems, and deployment platforms, making it suitable for projects of all sizes.

Jenkins Pipeline

A Jenkins Pipeline is a collection of automated stages that define the complete workflow for building, testing, and deploying an application. Pipelines enable organizations to implement the CI/CD process as code, ensuring consistency, repeatability, and easier maintenance of software delivery processes.

Advantages of Jenkins Pipeline

- Automates the entire software build, testing, and deployment process.
- Displays the workflow in clearly defined stages, making execution easier to monitor.

- Supports **Pipeline as Code**, allowing pipeline definitions to be stored and managed using version control systems.
- Enables advanced workflow capabilities such as parallel execution, conditional stage execution, and reusable pipeline scripts.

Types of Jenkins Pipelines

1. Declarative Pipeline

- Uses a structured and easy-to-understand syntax.
- Implemented inside a pipeline {} block containing stages and steps.
- Easier to develop, maintain, and debug, making it the preferred choice for most CI/CD projects.

2. Scripted Pipeline

- Written using Groovy syntax within a node {} block.
- Provides greater flexibility for implementing custom and complex workflows.
- Suitable for projects that require advanced scripting and highly customized pipeline logic.

Procedure

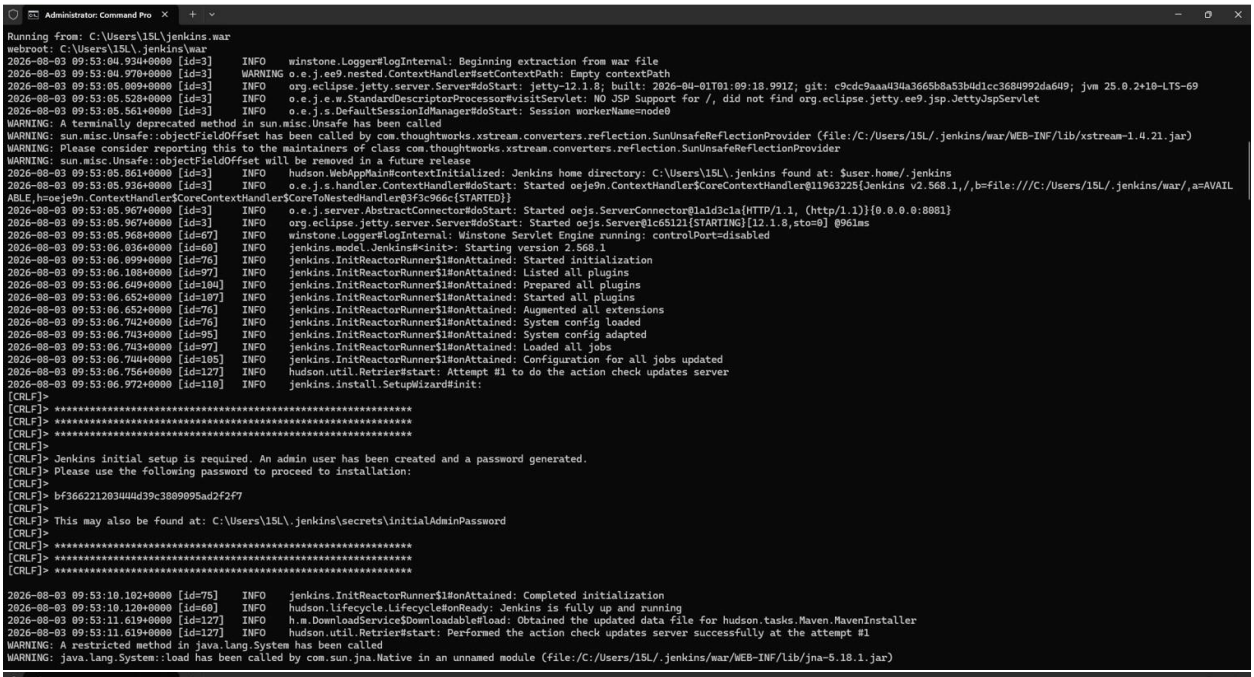
1. Verified that the Java Development Kit (JDK) was successfully installed and properly configured on the system.
2. Downloaded the jenkins.war file and launched the Jenkins server using the command:

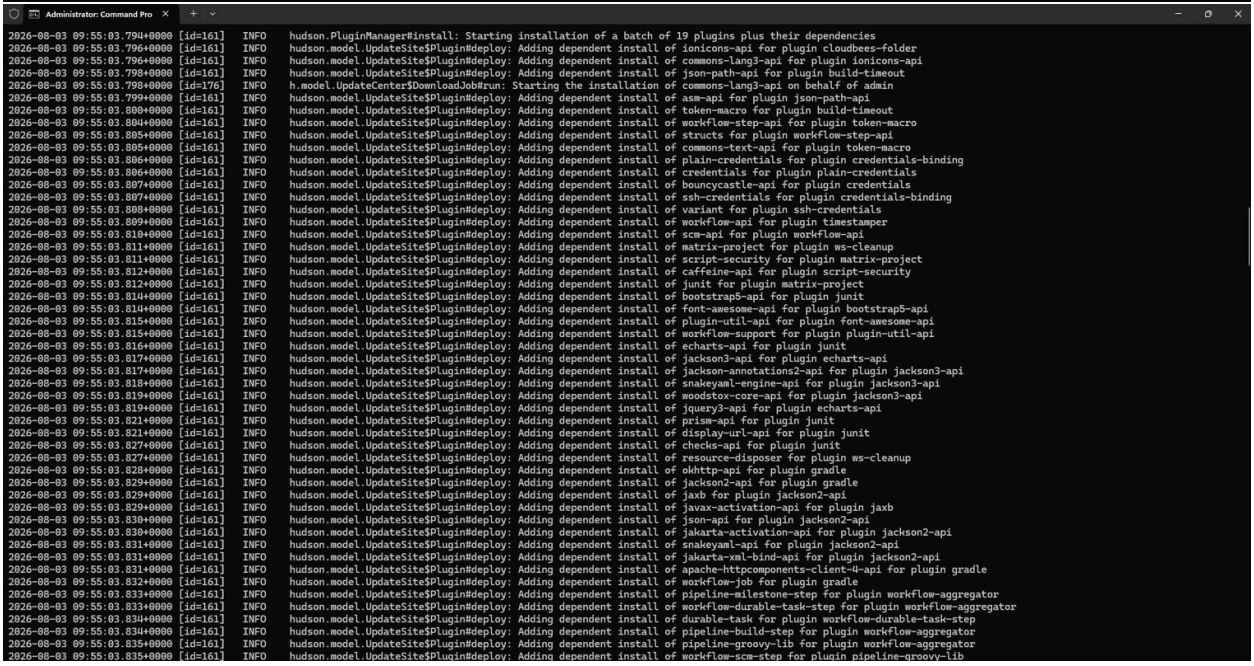
java -jar jenkins.war

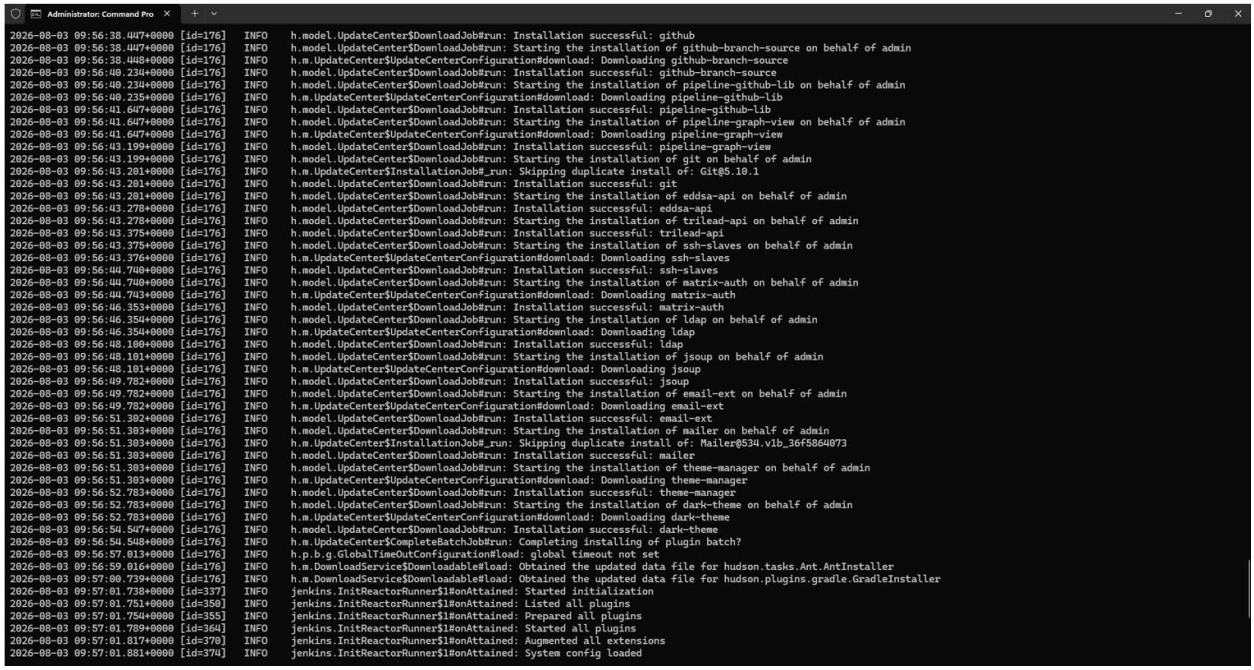
3. Accessed Jenkins through http://localhost:8080, entered the initial administrator password, installed the recommended plugins, and completed the administrator setup process.
4. Created two Jenkins Pipeline jobs:
    - A **Declarative Pipeline** using the pipeline {} syntax.
    - A **Scripted Pipeline** using the Groovy node {} syntax.

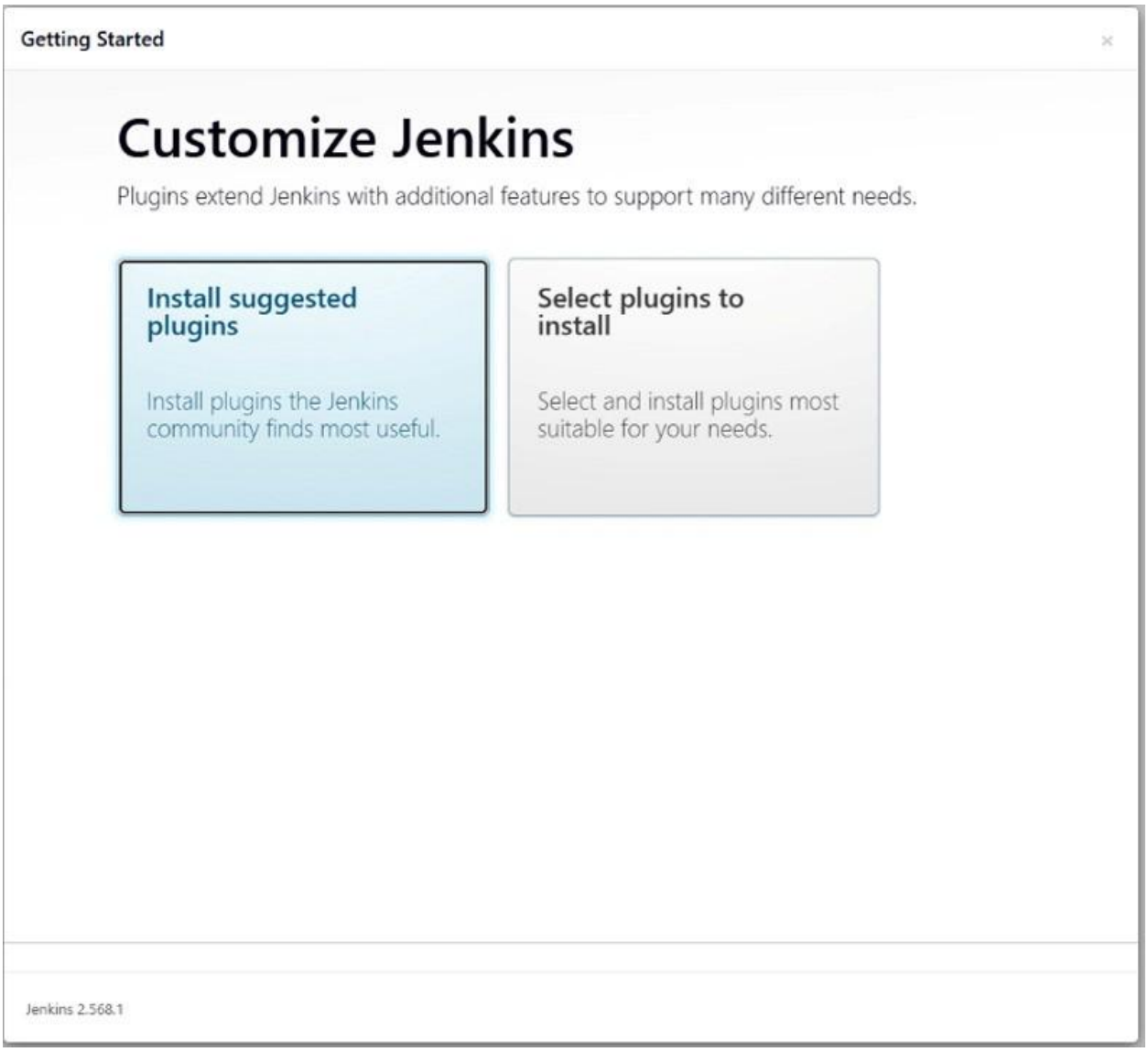
5. Executed both pipelines successfully and verified their completion by checking the build status and reviewing the Console Output in Jenkins.

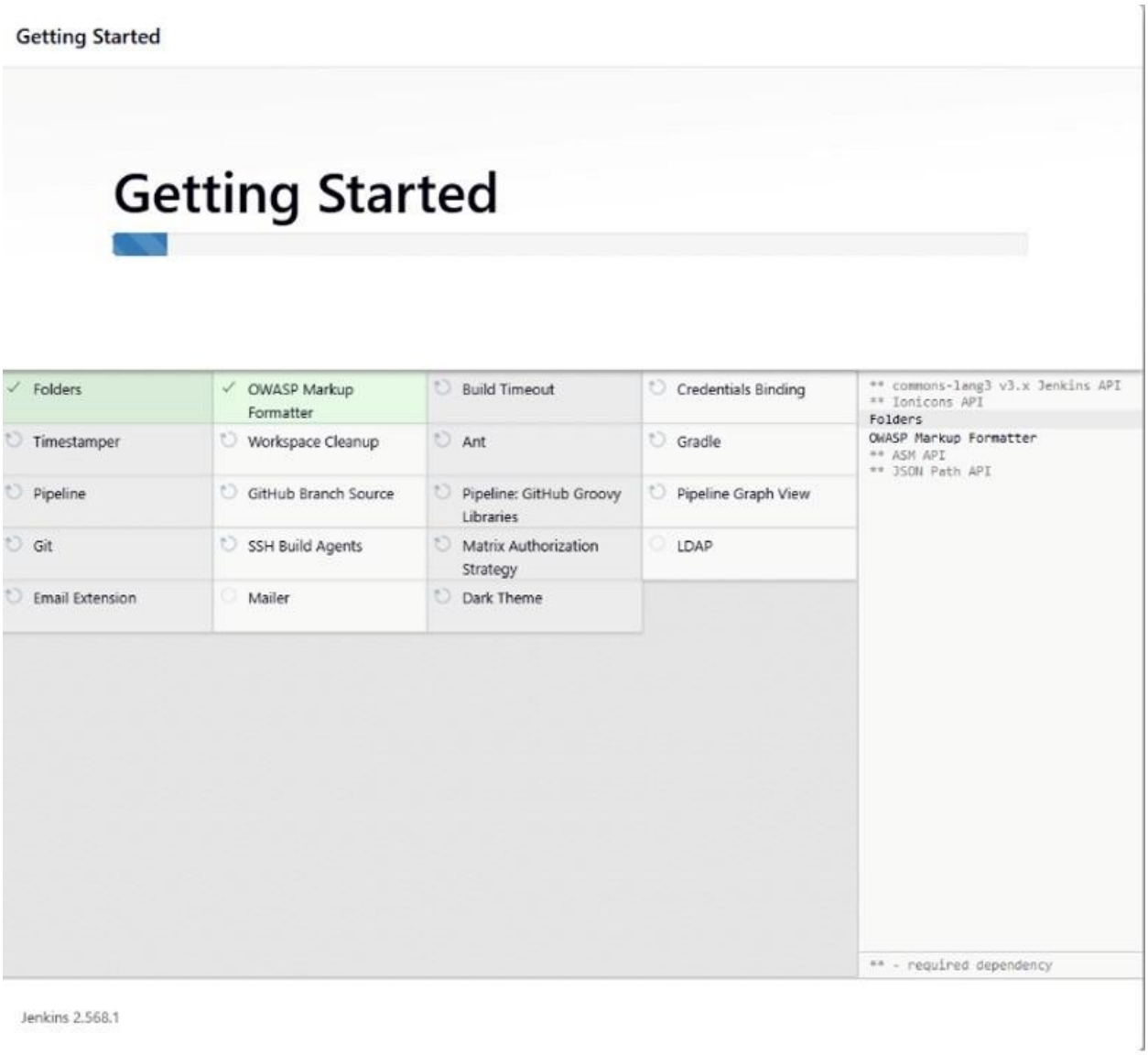
Output:

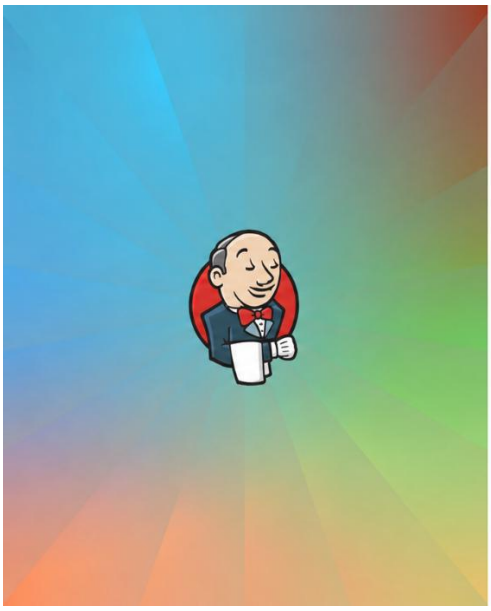


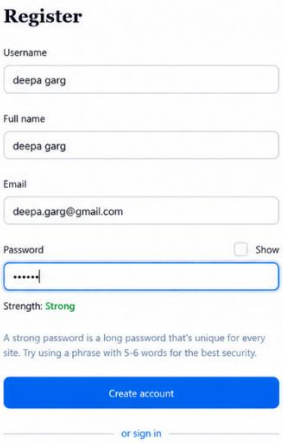


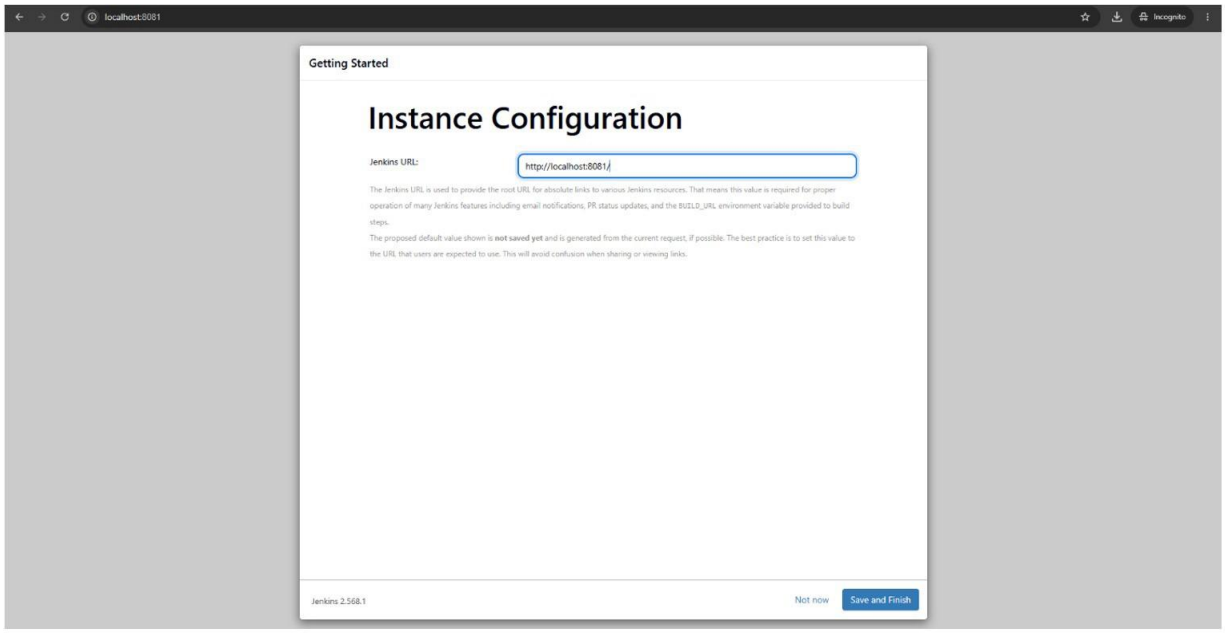

Administrator: Command Pro
2026-08-03 09:56:38.447+0000 [id=176] INFO h.model.UpdateCenter$DownloadJob#run: Installation successful: github
2026-08-03 09:56:38.447+0000 [id=176] INFO h.model.UpdateCenter$DownloadJob#run: Starting the installation of github-branch-source on behalf of admin
2026-08-03 09:56:38.448+0000 [id=176] INFO h.m.UpdateCenter$UpdateCenterConfiguration#download: Downloading github-branch-source
2026-08-03 09:56:40.234+0000 [id=176] INFO h.model.UpdateCenter$DownloadJob#run: Installation successful: github-branch-source
2026-08-03 09:56:40.234+0000 [id=176] INFO h.model.UpdateCenter$DownloadJob#run: Starting the installation of pipeline-github-lib on behalf of admin
2026-08-03 09:56:40.235+0000 [id=176] INFO h.m.UpdateCenter$UpdateCenterConfiguration#download: Downloading pipeline-github-lib
2026-08-03 09:56:41.647+0000 [id=176] INFO h.model.UpdateCenter$DownloadJob#run: Installation successful: pipeline-github-lib
2026-08-03 09:56:41.647+0000 [id=176] INFO h.model.UpdateCenter$DownloadJob#run: Starting the installation of pipeline-graph-view on behalf of admin
2026-08-03 09:56:41.647+0000 [id=176] INFO h.m.UpdateCenter$UpdateCenterConfiguration#download: Downloading pipeline-graph-view
2026-08-03 09:56:43.199+0000 [id=176] INFO h.model.UpdateCenter$DownloadJob#run: Installation successful: pipeline-graph-view
2026-08-03 09:56:43.199+0000 [id=176] INFO h.model.UpdateCenter$DownloadJob#run: Starting the installation of git on behalf of admin
2026-08-03 09:56:43.201+0000 [id=176] INFO h.m.UpdateCenter$InstallationJob#_run: Skipping duplicate install of: Git@5.10.1
2026-08-03 09:56:43.201+0000 [id=176] INFO h.model.UpdateCenter$DownloadJob#run: Installation successful: git
2026-08-03 09:56:43.201+0000 [id=176] INFO h.model.UpdateCenter$DownloadJob#run: Starting the installation of eddsa-api on behalf of admin
2026-08-03 09:56:43.278+0000 [id=176] INFO h.model.UpdateCenter$DownloadJob#run: Installation successful: eddsa-api
2026-08-03 09:56:43.278+0000 [id=176] INFO h.model.UpdateCenter$DownloadJob#run: Starting the installation of trilead-api on behalf of admin
2026-08-03 09:56:43.375+0000 [id=176] INFO h.model.UpdateCenter$DownloadJob#run: Installation successful: trilead-api
2026-08-03 09:56:43.375+0000 [id=176] INFO h.model.UpdateCenter$DownloadJob#run: Starting the installation of ssh-slaves on behalf of admin
2026-08-03 09:56:43.376+0000 [id=176] INFO h.m.UpdateCenter$UpdateCenterConfiguration#download: Downloading ssh-slaves
2026-08-03 09:56:44.740+0000 [id=176] INFO h.model.UpdateCenter$DownloadJob#run: Installation successful: ssh-slaves
2026-08-03 09:56:44.740+0000 [id=176] INFO h.model.UpdateCenter$DownloadJob#run: Starting the installation of matrix-auth on behalf of admin
2026-08-03 09:56:44.743+0000 [id=176] INFO h.m.UpdateCenter$UpdateCenterConfiguration#download: Downloading matrix-auth
2026-08-03 09:56:46.353+0000 [id=176] INFO h.model.UpdateCenter$DownloadJob#run: Installation successful: matrix-auth
2026-08-03 09:56:46.354+0000 [id=176] INFO h.model.UpdateCenter$DownloadJob#run: Starting the installation of ldap on behalf of admin
2026-08-03 09:56:46.354+0000 [id=176] INFO h.m.UpdateCenter$UpdateCenterConfiguration#download: Downloading ldap
2026-08-03 09:56:48.100+0000 [id=176] INFO h.model.UpdateCenter$DownloadJob#run: Installation successful: ldap
2026-08-03 09:56:48.101+0000 [id=176] INFO h.model.UpdateCenter$DownloadJob#run: Starting the installation of jsoup on behalf of admin
2026-08-03 09:56:48.101+0000 [id=176] INFO h.m.UpdateCenter$UpdateCenterConfiguration#download: Downloading jsoup
2026-08-03 09:56:49.782+0000 [id=176] INFO h.model.UpdateCenter$DownloadJob#run: Installation successful: jsoup
2026-08-03 09:56:49.782+0000 [id=176] INFO h.model.UpdateCenter$DownloadJob#run: Starting the installation of email-ext on behalf of admin
2026-08-03 09:56:49.782+0000 [id=176] INFO h.m.UpdateCenter$UpdateCenterConfiguration#download: Downloading email-ext
2026-08-03 09:56:51.302+0000 [id=176] INFO h.model.UpdateCenter$DownloadJob#run: Installation successful: email-ext
2026-08-03 09:56:51.303+0000 [id=176] INFO h.model.UpdateCenter$DownloadJob#run: Starting the installation of mailer on behalf of admin
2026-08-03 09:56:51.303+0000 [id=176] INFO h.m.UpdateCenter$InstallationJob#_run: Skipping duplicate install of: Mailer@534.v1b_36f5864073
2026-08-03 09:56:51.303+0000 [id=176] INFO h.model.UpdateCenter$DownloadJob#run: Installation successful: mailer
2026-08-03 09:56:51.303+0000 [id=176] INFO h.model.UpdateCenter$DownloadJob#run: Starting the installation of theme-manager on behalf of admin
2026-08-03 09:56:51.303+0000 [id=176] INFO h.m.UpdateCenter$UpdateCenterConfiguration#download: Downloading theme-manager
2026-08-03 09:56:52.783+0000 [id=176] INFO h.model.UpdateCenter$DownloadJob#run: Installation successful: theme-manager
2026-08-03 09:56:52.783+0000 [id=176] INFO h.model.UpdateCenter$DownloadJob#run: Starting the installation of dark-theme on behalf of admin
2026-08-03 09:56:52.783+0000 [id=176] INFO h.m.UpdateCenter$UpdateCenterConfiguration#download: Downloading dark-theme
2026-08-03 09:56:54.547+0000 [id=176] INFO h.model.UpdateCenter$DownloadJob#run: Installation successful: dark-theme
2026-08-03 09:56:54.548+0000 [id=176] INFO h.m.UpdateCenter$CompleteBatchJob#run: Completing installing of plugin batch?
2026-08-03 09:56:57.813+0000 [id=176] INFO h.p.b.g.GlobalTimeOutConfiguration#load: global timeout not set
2026-08-03 09:56:59.016+0000 [id=176] INFO h.m.DownloadService$Downloadable#load: Obtained the updated data file for hudson.tasks.Ant.AntInstaller
2026-08-03 09:57:00.739+0000 [id=176] INFO h.m.DownloadService$Downloadable#load: Obtained the updated data file for hudson.plugins.gradle.GradleInstaller
2026-08-03 09:57:01.738+0000 [id=337] INFO jenkins.InitReactorRunner$1#onAttained: Started initialization
2026-08-03 09:57:01.751+0000 [id=350] INFO jenkins.InitReactorRunner$1#onAttained: Listed all plugins
2026-08-03 09:57:01.754+0000 [id=355] INFO jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
2026-08-03 09:57:01.789+0000 [id=364] INFO jenkins.InitReactorRunner$1#onAttained: Started all plugins
2026-08-03 09:57:01.817+0000 [id=370] INFO jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
2026-08-03 09:57:01.881+0000 [id=374] INFO jenkins.InitReactorRunner$1#onAttained: System config loaded

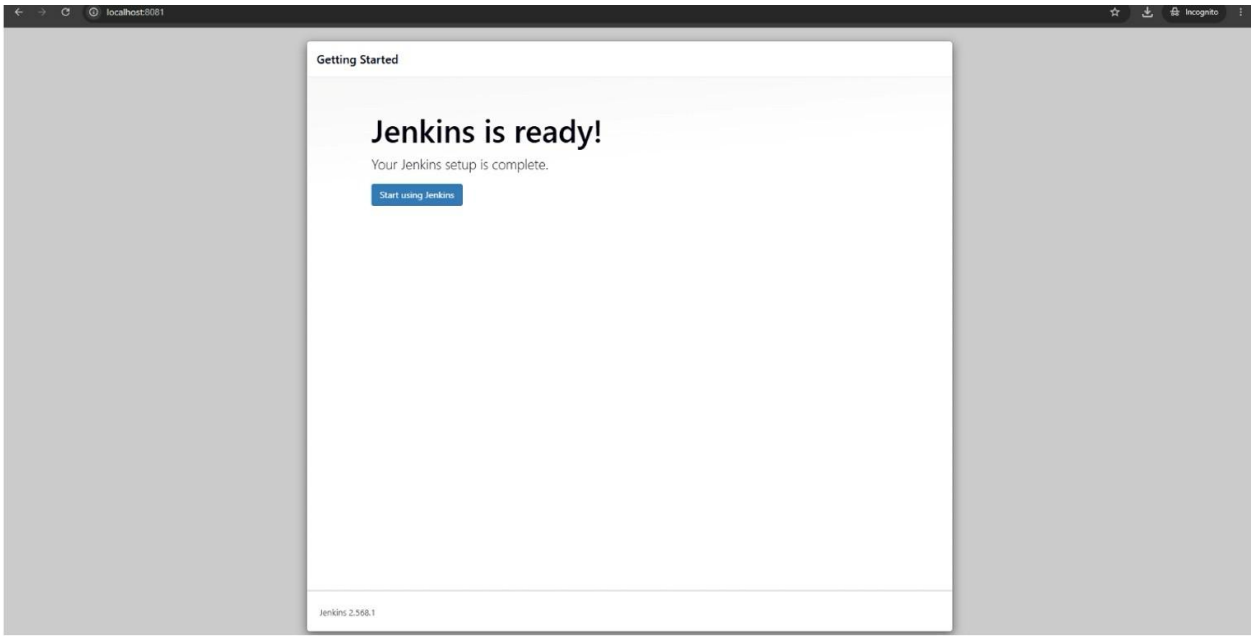

Getting Started
Customize Jenkins
Plugins extend Jenkins with additional features to support many different needs.
Install suggested plugins
Install plugins the Jenkins community finds most useful.
Select plugins to install
Select and install plugins most suitable for your needs.
Jenkins 2.568.1

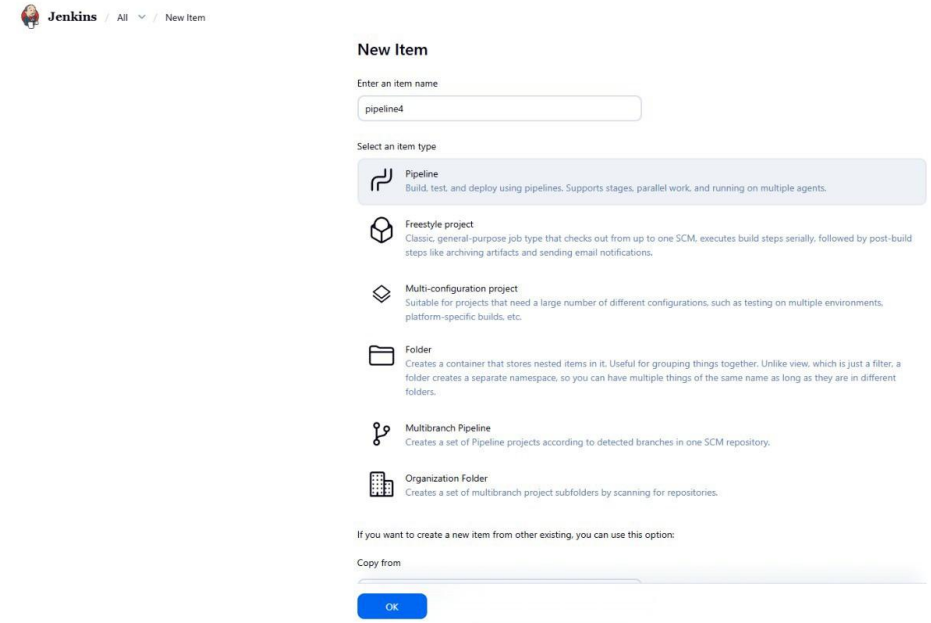

Getting Started
Getting Started
Folders
OWASP Markup Formatter
Build Timeout
Credentials Binding
Timestamper
Workspace Cleanup
Ant
Gradle
Pipeline
GitHub Branch Source
Pipeline: GitHub Groovy Libraries
Pipeline Graph View
Git
SSH Build Agents
Matrix Authorization Strategy
LDAP
Email Extension
Mailer
Dark Theme
** commons-lang3 v3.x Jenkins API
** Ionicons API
Folders
OWASP Markup Formatter
** ASM API
** JSON Path API
** - required dependency
Jenkins 2.568.1

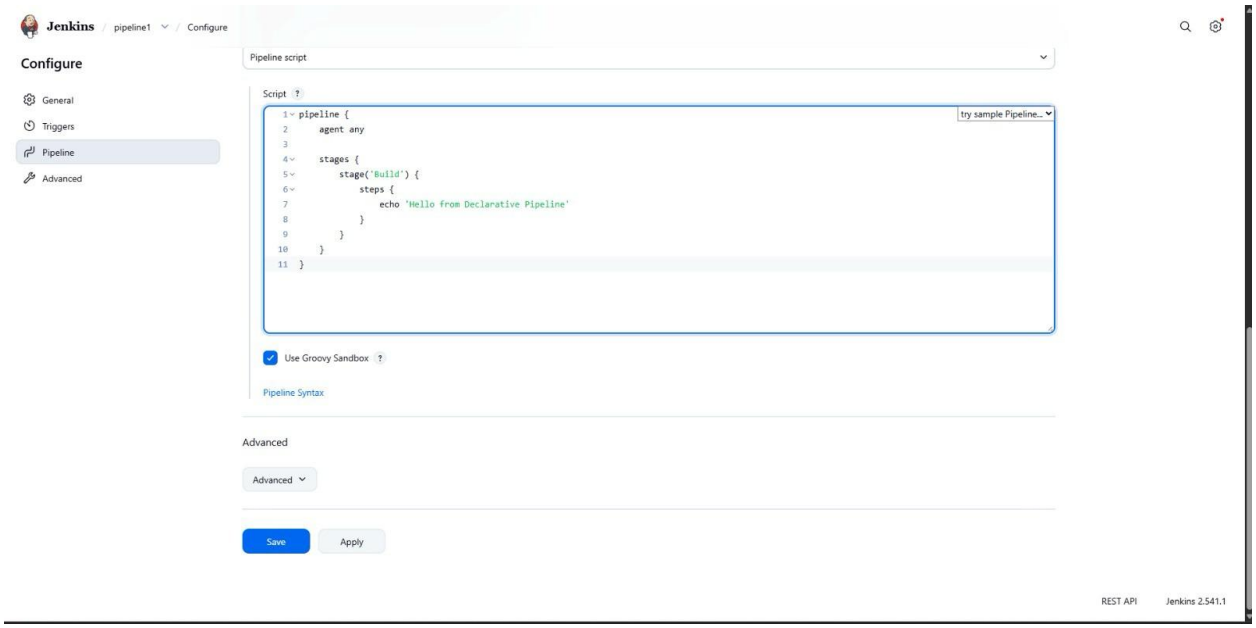



Register
Username
deepa garg
Full name
deepa garg
Email
deepa.garg@gmail.com
Password
Show
Strength: Strong
A strong password is a long password that's unique for every site. Try using a phrase with 5-6 words for the best security.
Create account
or sign in

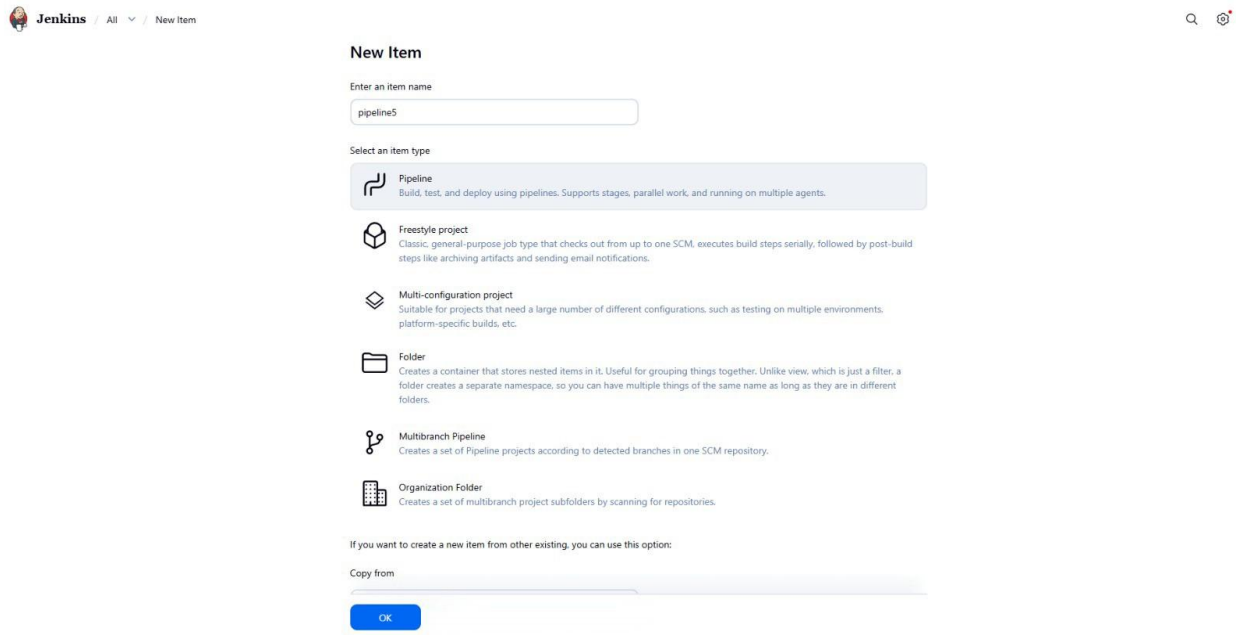

localhost:8081
Incognito
Getting Started
Instance Configuration
Jenkins URL:
http://localhost:8081/
The Jenkins URL is used to provide the root URL for absolute links to various Jenkins resources. That means this value is required for proper operation of many Jenkins features including email notifications, PR status updates, and the BUILD_URL environment variable provided to build steps.
The proposed default value shown is not saved yet and is generated from the current request, if possible. The best practice is to set this value to the URL that users are expected to use. This will avoid confusion when sharing or viewing links.
Jenkins 2.568.1
Not now
Save and Finish

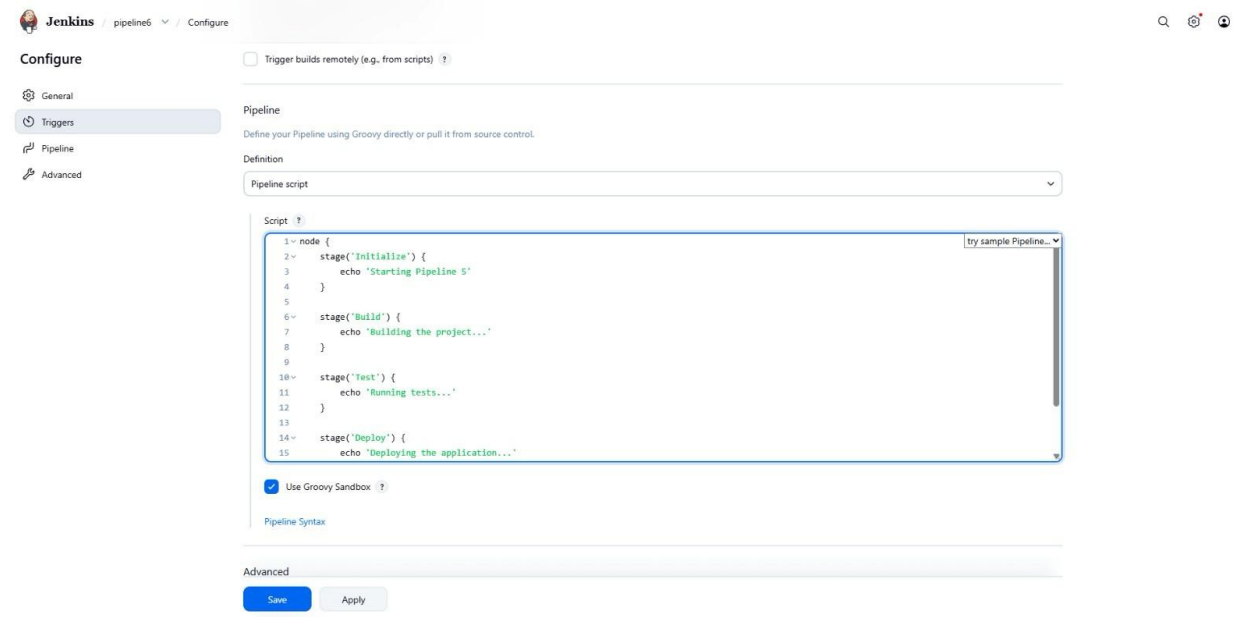

localhost:8081
Incognito
Getting Started
Jenkins is ready!
Your Jenkins setup is complete.
Start using Jenkins
Jenkins 2.568.1

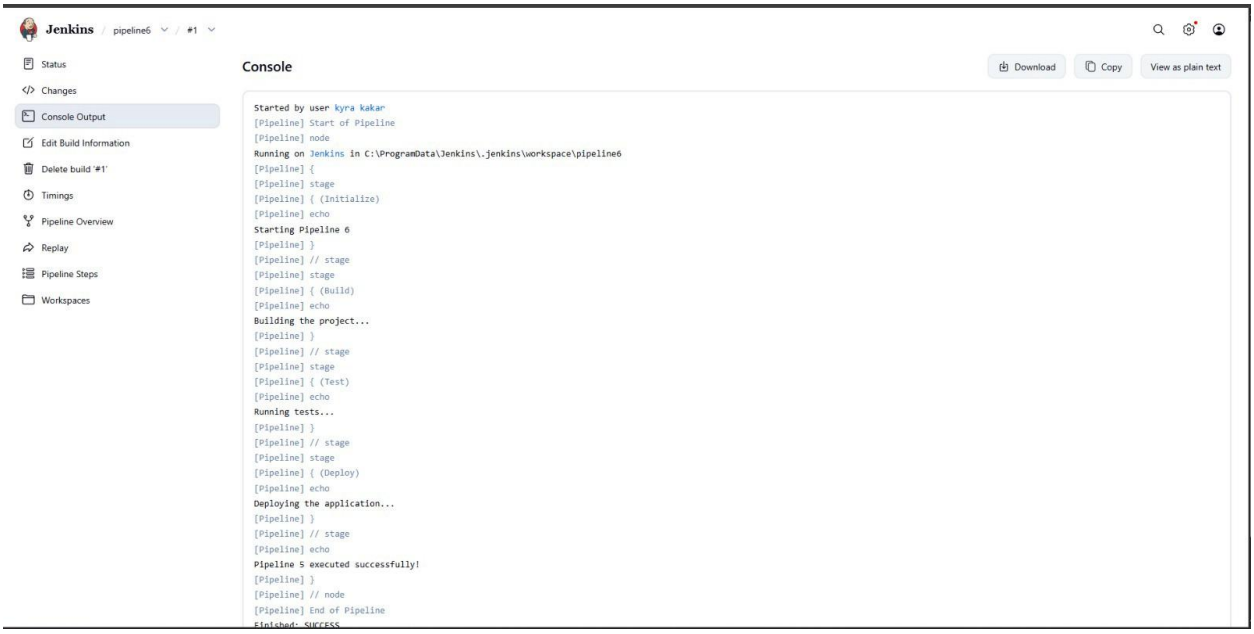

Jenkins / All / New Item
New Item
Enter an item name
pipeline4
Select an item type
Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
Multi-configuration project
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
Folder
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.
Multibranch Pipeline
Creates a set of Pipeline projects according to detected branches in one SCM repository.
Organization Folder
Creates a set of multibranch project subfolders by scanning for repositories.
If you want to create a new item from other existing, you can use this option:
Copy from
OK


Jenkins / pipeline1 / Configure
Configure
General
Triggers
Pipeline
Advanced
Pipeline script
Script
try sample Pipeline...
pipeline {
agent any
stages {
stage('Build') {
steps {
echo 'Hello from Declarative Pipeline'
}
}
}
}
Use Groovy Sandbox
Pipeline Syntax
Advanced
Advanced
Save
Apply
REST API
Jenkins 2.541.1


Jenkins / pipeline4 / #1
Status
Changes
Console Output
Edit Build Information
Delete build '#1'
Timings
Pipeline Overview
Replay
Pipeline Steps
Workspaces
Console
Download
Copy
View as plain text
Started by user kyra kakar
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\pipeline4
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] echo
Hello from Scripted Pipeline
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS


Jenkins / All / New Item
New Item
Enter an item name
pipeline5
Select an item type
Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
Multi-configuration project
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
Folder
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.
Multibranch Pipeline
Creates a set of Pipeline projects according to detected branches in one SCM repository.
Organization Folder
Creates a set of multibranch project subfolders by scanning for repositories.
If you want to create a new item from other existing, you can use this option:
Copy from
OK


Jenkins / pipeline6 / Configure
Configure
General
Triggers
Pipeline
Advanced
Trigger builds remotely (e.g., from scripts)
Pipeline
Define your Pipeline using Groovy directly or pull it from source control.
Definition
Pipeline script
Script
try sample Pipeline...
node {
stage('Initialize') {
echo 'Starting Pipeline 5'
}
stage('Build') {
echo 'Building the project...'
}
stage('Test') {
echo 'Running tests...'
}
stage('Deploy') {
echo 'Deploying the application...'
Use Groovy Sandbox
Pipeline Syntax
Advanced
Save
Apply


Jenkins / pipeline6 / #1
Status
Changes
Console Output
Edit Build Information
Delete build '#1'
Timings
Pipeline Overview
Replay
Pipeline Steps
Workspaces
Console
Download
Copy
View as plain text
Started by user kyra kakar
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\pipeline6
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Initialize)
[Pipeline] echo
Starting Pipeline 6
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] echo
Building the project...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Test)
[Pipeline] echo
Running tests...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy)
[Pipeline] echo
Deploying the application...
[Pipeline] }
[Pipeline] // stage
[Pipeline] echo
Pipeline 5 executed successfully!
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS



Conclusion:

This experiment helped us understand the basics of Continuous Integration (CI) and Continuous Delivery/Deployment (CD) using Jenkins. We installed and configured Jenkins, explored its interface, and successfully created and executed both Declarative and Scripted Pipelines.

We learned how Jenkins automates software development tasks such as building, testing, and deployment through Pipeline as Code, making the CI/CD process faster, more reliable, and easier to manage.

Overall, the experiment provided practical knowledge of Jenkins and strengthened our understanding of automated CI/CD workflows used in modern DevOps practices.

LO3 mapped successfully.